Software Design and Architecture

CCSW 311

Project

[SE3]

Group 3

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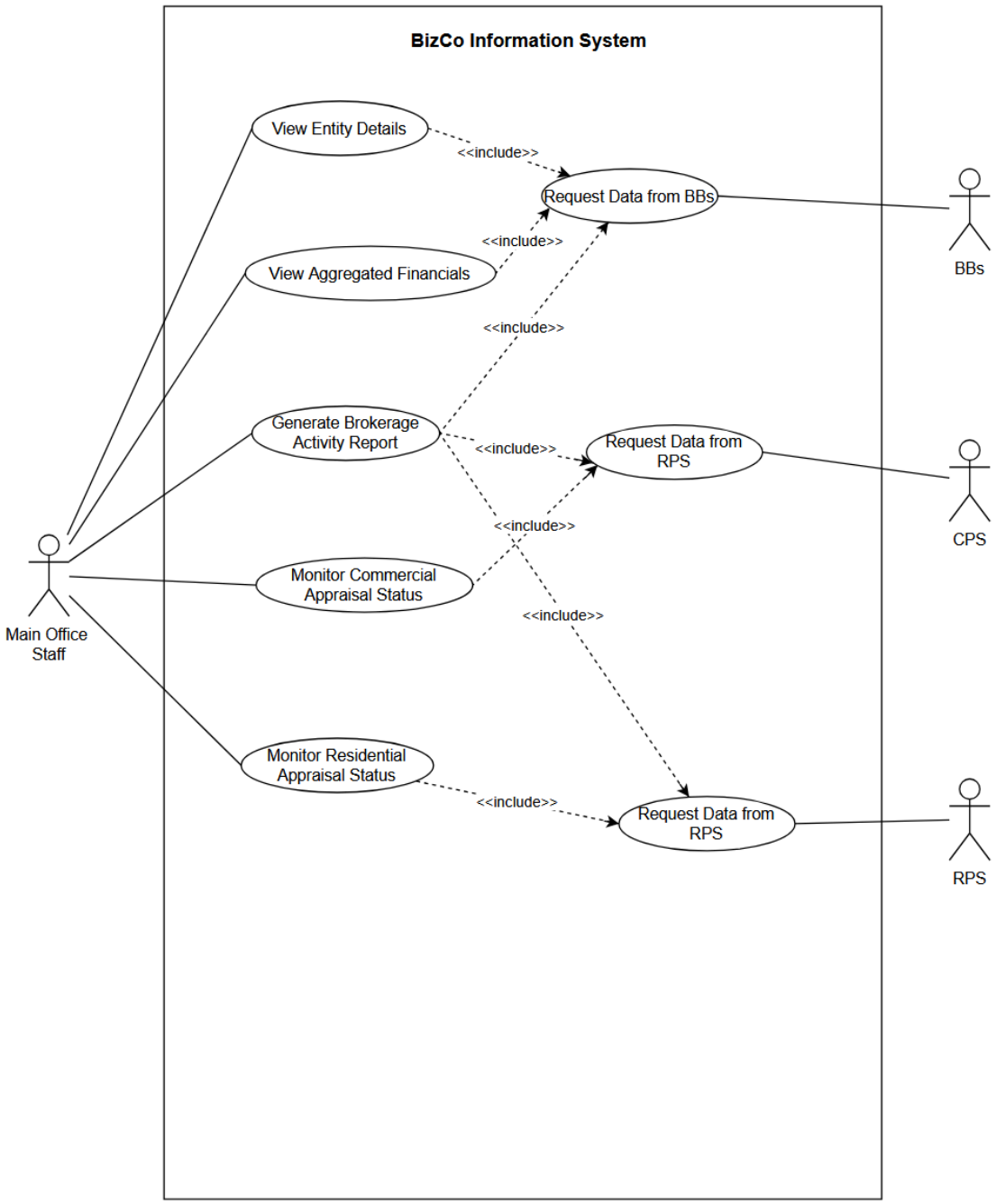
General Analysis

i. Use Case Diagram:



Use Case Description Table:

ID	Name	Description
UC-1	View Entity Details	Allows Main Office Staff to retrieve and display aggregated contact, address, contract terms, and service agreement details for all organizational entities.
UC-2	View Aggregated Financials	Enables the user to generate and view a consolidated report of accounts receivable data gathered from all branch offices.
UC-3	Generate Brokerage Activity Report	Allows the user to compile and display a comprehensive report of brokerage activities, filtering and aggregating data across branches, lenders, and appraisers.
UC-4	Monitor Commercial Appraisal Status	Enables the user to view the status (submitted, assigned, fulfilled) of commercial property appraisal requests across all branches.
UC-5	Monitor Residential Appraisal Status	Enables the user to view the status (submitted, assigned, fulfilled) of residential property appraisal requests across all branches.
UC-6	Request Data from BBS	The system initiates a standardized request to the Brokerage Billing System to retrieve all financial, contract, and entity contact data.
UC-7	Request Data from CPS	The system initiates a standardized request to the Commercial Property System to retrieve commercial appraisal request status and details.
UC-8	Request Data from RPS	The system initiates a standardized request to the Residential Property System to retrieve residential appraisal request status and details.

ii. Business Goals, Use cases, Constraints

Business Goals (BG):

ID	Business Goals
BG-1	Establish BizCo as the industry leader in brokering appraisal for commercial and residential properties
BG-2	Increase market share of brokerage services
BG-3	Dramatically decrease response time to changing market conditions
BG-4	Have direct, secure access to brokerage information 24/7
BG-5	Have reporting that aggregates, collates, and communicates timely information clearly and as needed

Use Case (UC):

ID	Use case
UC-1	View Entity Details
UC-2	View Aggregated Financials
UC-3	Generate Brokerage Activity Report
UC-4	Monitor Commercial Appraisal Status
UC-5	Monitor Residential Appraisal Status
UC-6	Request Data from BBS
UC-7	Request Data from CPS
UC-8	Request Data from RPS

Constraint (CN):

ID	Constraint
CN-1	The BizCo systems architect has designed a Layered Architecture
CN-2	Must integrate data from three standalone systems that are not integrated withing the branch and does not integrate externally
CN-3	Overcome the limitations of generating biweekly and hard copy reports that must be sent to main office.

Identify the quality attribute requirements.

i. ASR's and quality attributes scenarios

ASR-01 (Performance): The BIS must display any aggregated reports (e.g., all branch activity) for 100 concurrent main office staff users within 5 seconds during peak usage.

Key Part	Description
Source of stimulus	Main office staff
Stimulus	Staff initiates the view all branch activity (UC-05)
Environment	During peak hours
Artifact	Messaging infrastructure
Response	Display the report successfully
Response measure	Display time is within 5 seconds
Origin	BG-05

ASR-02 (Availability): If a single branch system is offline, the BIS must continue to display the last known data without failure.

Key Part	Description
Source of stimulus	External software
Stimulus	Fault crash in a single branch system
Environment	Normal operation
Artifact	Central data repository
Response	Continue to display the last known data
Response measure	Without failure
Origin	BG-03

ASR-03 (Security): The system shall prevent any unauthorized user from viewing sensitive data and logs all denial attempts.

Key Part	Description
Source of stimulus	Unauthorized user
Stimulus	Attempt to access sensitive data
Environment	Normal operation
Artifact	System data
Response	Logs all denial attempts
Response measure	zero unauthorized access per month
Origin	BG-04

ASR-04 (Scalability): The messaging layer must be able to successfully handle a sustained event from branch staff users when they upload or retrieve data with a throughput of 200 events per second during peak usage period.

Key Part	Description
Source of stimulus	Branch staff users
Stimulus	Upload or retrieve data
Environment	Peak usage period
Artifact	Messaging infrastructure
Response	Successfully handle sustained event
Response measure	With a throughput of 200 events per second
Origin	BG-01

ASR-05 (Modifiability): The Messaging Infrastructure must be easily modified to accept new types of appraisal or billing data from the branch systems.

Key Part	Description
Source of stimulus	Developer
Stimulus	accept new types of appraisal or billing data
Environment	Development time
Artifact	Messaging infrastructure
Response	Modified changes are implemented, deployed and active
Response measure	In under 30 minutes without downtime
Origin	BG-03

ASR-06 (Maintainability): The Messaging Infrastructure must allow system administrators to identify, isolate, and restart a failing messaging component without interrupting message flow for other branch systems.

Key Part	Description
Source of stimulus	System administrators
Stimulus	A messaging component fails or behaves abnormally and requires maintenance.
Environment	Normal operation
Artifact	Messaging infrastructure
Response	The failing component is identified, isolated and restarted.
Response measure	Completed within 10 minutes with zero message loss
Origin	CN-1

ASR-07 (Interoperability): The Messaging infrastructure must successfully exchange appraisal-related and billing-related messages with three standalone subsystems that each use different data formats and protocols.

Key Part	Description
Source of stimulus	Three standalone subsystems
Stimulus	Sending appraisal or billing messages with different formats/protocols
Environment	Normal operation
Artifact	Messaging infrastructure
Response	Translate and process all incoming and outgoing messages successfully.
Response measure	100% compatibility with all three standalone subsystems
Origin	CN-2

ASR-08 (Reliability): The messaging infrastructure must ensure that all submitted messages are delivered only once even when a network interruption occurs between the subsystems and IBS.

Key Part	Description
Source of stimulus	Subsystems
Stimulus	Network interruption between subsystem and IBs
Environment	Normal operation
Artifact	Messaging infrastructure
Response	Retrieve and deliver message only once when network resumes.
Response measure	0% message duplication and 0% message loss across 1000 transmissions
Origin	BG-4

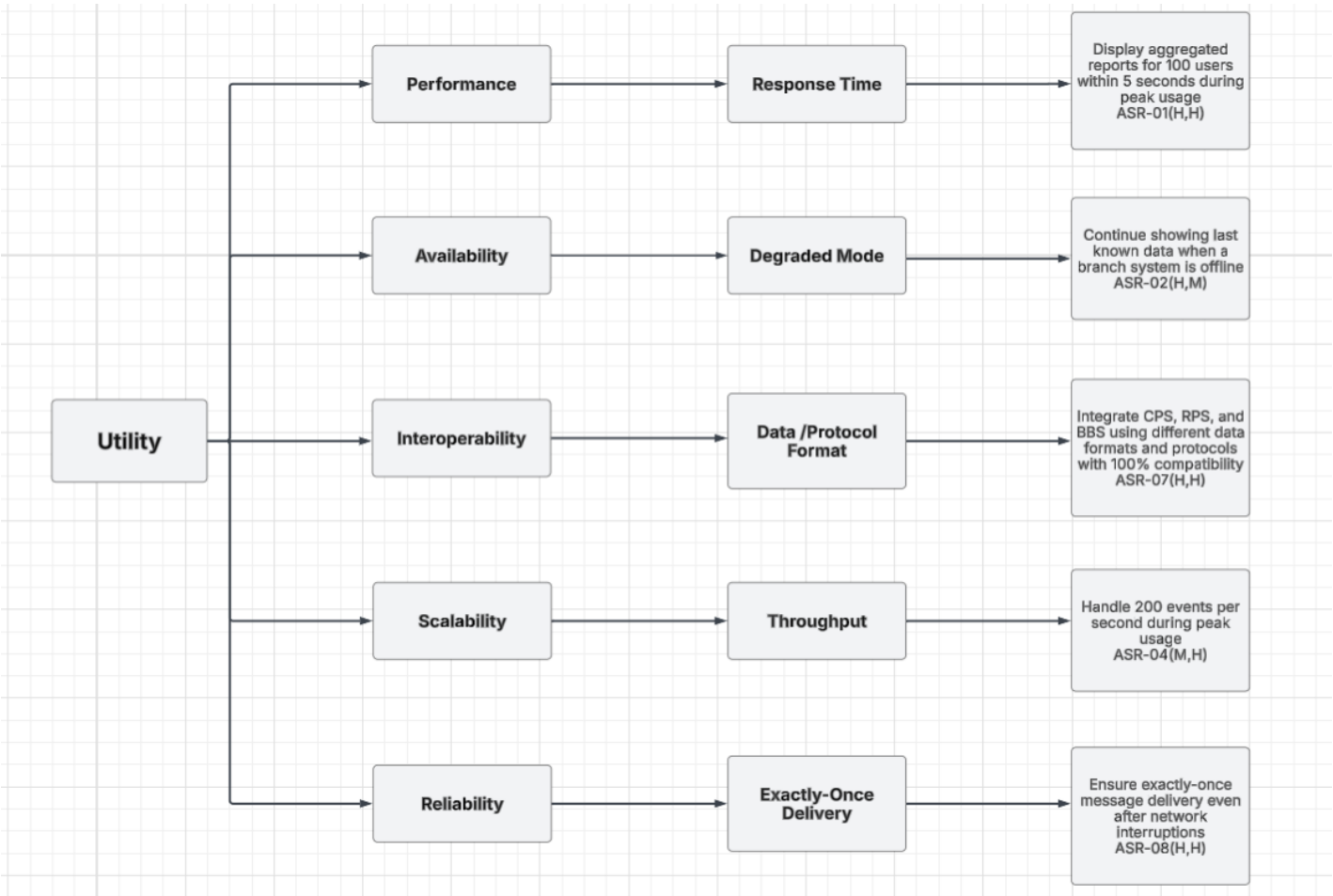
ASR-09 (Portability): The Messaging Infrastructure must be deployable on different operating environments (e.g., Windows Server, Linux-based servers, or cloud containers) without requiring architectural changes.

Key Part	Description
Source of stimulus	Deployment engineer
Stimulus	Deploy the messaging infrastructure on a new server
Environment	Scaling scenario
Artifact	Messaging infrastructure
Response	Deploys successfully with only configuration adjustments
Response measure	Deployment completed within **1 hour** with no code modifications
Origin	BG-3

ASR-10 (Usability): The Messaging Infrastructure's API must be easy for developers to understand and use, enabling branch system developers to send or retrieve messages with minimal effort.

Key Part	Description
Source of stimulus	Branch system developer
Stimulus	Integrate with messaging infrastructure API
Environment	Development time
Artifact	Messaging infrastructure API
Response	Developer is able to complete message integration tasks easily
Response measure	API integration completed within 2 hours using only the provided documentation
Origin	CN-2

ii. Utility Tree:



iii. Quality attributes

Quality Attribute	Origin	source	Value impact	Justification
Performance	BG-05	ARS-01	H	Fast access to aggregated report is essential for supporting BG-5 as main office staff rely on timely information to make informed decision if the report cannot display within 5 seconds the information becomes less useful and does not meet the requirements for timely and clear reporting
			H	Meeting a 5 second limit for large, aggregated reports with 100 concurrent users at peak time requires architectural decisions for data access messaging flow and possible caching affects how the system is designed
Availability	BG-3	ARS-02	H	Users still need to see information even if one branch system crashes this support continuous operation and helps the company react to market conditions without waiting for the branch to come back
			M	Showing “last known data” can be done using caching or stored snapshots this needs support in the architecture, but it does not require a full redesign of the messaging infrastructure
Scalability	BG-1	ARS-04	M	Scalability has medium business value because handling 200 events per second mainly supports future growth and increased branch activity rather than being a strict requirement for basic system operation. The system can still function correctly under lower loads.
			H	Supporting this throughput directly impacts the messaging architecture it requires scalable queues concurrency handling and possible load balancing
Interoperability	CN-2	ARS-07	H	The BIS exists to integrate CPS, RPS, and BBS if the messaging infrastructure cannot talk to all three with full compatibility the main purpose of the system fails
			H	Supporting three standalone subsystems that use different

				data format and protocols requires significant architecture work this affects the core structure of the messaging layer
Reliability	BG-4	ARS-08	H	Reliable message delivery is essential for billing contracts and appraisal results any lost or duplicated messages can cause financial and business errors
			H	Delivering each message only once even after a network interruption require important changes in how the messaging system is built because this behavior must be always guaranteed

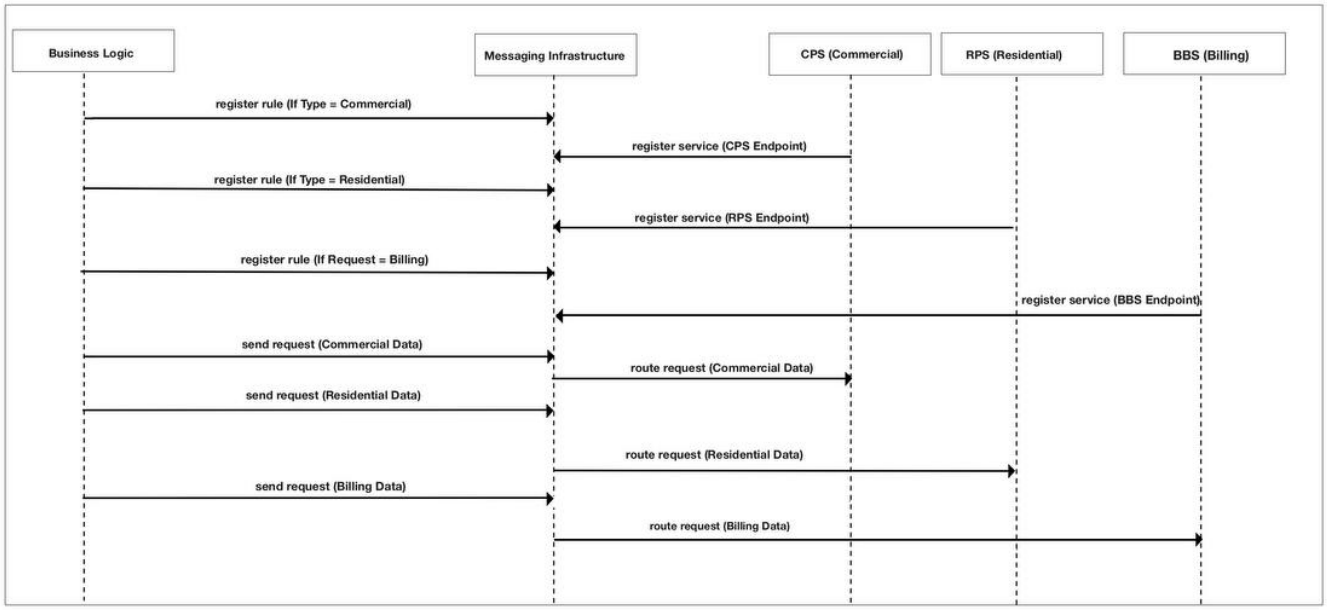
iv. Prioritize Quality attribute.

	Quality Attribute	Justification (method of prioritization)
1	Interoperability	Because BIS cannot function unless CPS, RPS, and BBS communicate correctly all data reporting depends on successful integration of the three standalone systems (CPS, RPS, BBS) is a mandatory prerequisite for any BIS functionality. It is prioritized first because it has both high business value and high architectural impact in the Utility Tree.
2	Performance	Fast access to aggregated reports is essential to support BG-5 and enable timely decision making. Achieving less than 5 seconds response for 100 users strongly affects architecture choices. It is ranked second because it also has high value and high impact, but interoperability is more foundational to system functionality.
3	Reliability	Exactly once delivery is critical for financial accuracy in billing and appraisal results any message loss or duplication creates business errors and ensuring delivery requires architecture mechanisms. It is periodized third because it has high value and high impact but comes after performance and interoperability in foundational importance.
4	Scalability	The system must handle 200 events during peak usage to support business growth and additional branches this has a significant impact on messaging throughput and concurrency design. It is given fourth because its ASR has high architectural impact but is slightly less business critical than the top three.
5	Availability	The main office must continue accessing brokerage data even if a branch system fails this supports business continuity but has lower architectural impact. It is ranked fifth because its ASR has high value but medium impact making it less critical than attributes above.

Messaging Infrastructure Layer

Based on the description provided, select and illustrate the architectural pattern that best satisfies the messaging infrastructure requirements. Then, instantiate the selected pattern by:

Dynamic Routing Pattern (SOA)



i. roles and responsibilities

Sender (Business Logic Layer): This participant acts as the service consumer. Its role is to initiate business requests (such as submitting an appraisal request or querying financial data) and send them to the messaging infrastructure. It is completely decoupled by the receivers, meaning it operates without knowledge of the specific location, identity, or data format of the legacy systems.

Dynamic Router (Messaging Infrastructure): This is the central component of the pattern. Its primary responsibility is to inspect the content of incoming messages (e.g., checking if the property type is "Commercial", "Residential", or if the request is for "Billing and Contract") and apply specific routing rules to forward the message to the correct destination system (CPS, RPS, or BBS). It also handles data transformation to ensure interoperability between the new BIS and the legacy systems.

Receivers (Legacy Systems - CPS, RPS, BBS): These participants are the existing standalone service providers. Their role is to receive routed messages, process the specific business transactions (such as fulfilling a commercial appraisal in CPS, a residential one in RPS, processing billing and Service Agreements in BBS)and return the requested data or confirmation to the router.

ii. relationships and interactions

Decoupled Interaction: The relationship between the Sender (Business Logic) and the Receivers (Legacy Systems) is completely decoupled. The Sender does not know the location, API definition, or existence of the specific legacy systems. It only interacts with the Dynamic Router interface.

Hub-and-Spoke Relationship: The interactions follow a hub-and-spoke model where the Messaging Infrastructure (Router) acts as the central hub. All messages from the Business Logic layer must pass through this hub, which then creates individual interactions with the spoke systems (CPS, RPS, BBS) based on the routing logic.

Asynchronous Communication: The interaction mode is asynchronous. The Business Logic layer sends the request message to the Messaging Infrastructure and can continue processing other tasks without waiting for an immediate connection to the legacy systems. The Router handles the delivery and response collection in the background.

Rule-Based Interaction: The interaction between the Router and the Receivers is governed by dynamic rules (e.g., If property type = Commercial, THEN route to CPS). This allows the system to change behavior by updating rules without changing the code of the sender or receiver

iii. How they collaborate to achieve the objectives.

To achieve the system's integration goals (specifically satisfying Interoperability ASR-07), the participants collaborate in the following sequence:

Request Initiation: The Business Logic Layer sends a unified "Appraisal Request" message to the Messaging Infrastructure. This message contains the business data (e.g., property details) without specifying which legacy system should handle it.

Message Inspection & Routing: The Dynamic Router intercepts the message and inspects its content. It evaluates the defined routing rules against the message data. For instance, it checks the property_type field or request context:

- IF the type is "Commercial", the router selects the CPS channel.
- IF the type is "Residential", the router selects the RPS channel.
- IF the request is for "Billing", the router selects the BBS channel.

Transformation & Forwarding: Before forwarding, the Router transforms the message into the specific data format required by the selected Legacy System (solving the integration challenge). It then forwards the request asynchronously to the correct endpoint.

Service Execution: The selected Legacy System (Receiver) processes the request using its internal logic and generates the result.

Response Delivery: The Legacy System returns the result to the Router. The Router normalizes this response into a standard format and sends it back to the Business Logic Layer, completing the integration loop.

iv. Pattern tradeoffs

Pattern	Dynamic Routing Pattern (SOA)
Overview	A central Message Router acts as an intermediary between the Business Logic and legacy systems. It inspects message content (property type or request context) and applies rules to route the request to the correct destination service (CPS, RPS, or BBS), ensuring seamless integration without the sender knowing the details of the receivers
Elements	1. Sender: Business Logic Layer (Initiates requests). 2. Dynamic Router: Messaging Infrastructure (Inspects and routes messages). 3. Receivers: Legacy Systems: CPS (Commercial), RPS (Residential), and BBS (Billing and Contracts).
Relations	Decoupled: The Sender interacts only with the Router, not the Receivers. Asynchronous: The Router handles message delivery in the background. Rule-Based: Routing decisions are driven by dynamic rules (e.g., If Type=Commercial -> Route to CPS).
Constraints	1. Data Transformation: The router must be capable of converting data into the specific formats required by each legacy system. 2. Rule Consistency: Routing rules must be precisely defined to avoid logic conflicts or "dropped" messages.

Tradeoffs	Benefits (Pros): • High Interoperability (ASR-07): Connects disparate legacy systems effectively. • Maintainability: Adding a new branch system only requires updating router rules, not the Business Logic. Liabilities (Cons): • Performance Overhead (ASR-01): Inspecting every message adds slight latency. • Single Point of Failure: If the Router goes down, the entire integration stops

Applying Tactics

i. Security

Security is important because BIS handles sensitive financial data including account receivables, contracts, and billing information. Unauthorized access could cause financial losses and damage BizCo's reputation.BG-04 requires direct and secure access to brokerage information 24/7 Since BIS connects three standalone systems across multiple branch offices to a central main office, the system has many potential entry points for attacks

Improving Security protects the other quality attributes. High Performance and Availability are useless if unauthorized users can access or corrupt data. While Interoperability and Reliability ensure smooth communication between systems, Security prevents that communication from being intercepted or modified by attackers

ii. Security tactics to be applied:

Detect Intrusion (Detect Attacks)

This tactic monitors network traffic and compares it to known attack patterns. The messaging layer checks incoming messages from branch systems (CPS, RPS, BBS) for suspicious activities like unusual protocols, strange data sizes, or unknown source addresses. When suspicious messages are detected, they are flagged for review.
This supports ASR-03's requirement to prevent unauthorized access.

Authenticate Actors (Resist Attacks)

This tactic verifies that users and systems are who they claim to be using passwords, certificates, or tokens. Branch systems must prove their identity before sending messages, and main office staff must authenticate before viewing reports. Ensuring only legitimate entities can use the BIS.

Revoke Access (React to Attacks)

This tactic blocks access when attacks are detected. If the system detects repeated failed login attempts or unusual message patterns, it can temporarily block the suspicious user or system. For example, if a branch system is compromised and sends corrupt data, it can be blocked until fixed. This limits damage from ongoing attacks while allowing users to continue working.

Maintain Audit Trail (Recover from Attacks)

This tactic records all system activities including who accessed what, when they accessed it, and what they did. The audit trail logs every report access, every message from branch systems, and every authentication attempt. This record helps trace attacker actions after an incident and supports ASR-03's requirement to log all access denials.
It also provides evidence for accountability and compliance.

iii. Tradeoffs:

Slower Performance

Checking every message for intrusions and authenticating every user takes time. If authentication takes 100ms and intrusion detection takes 50ms per message, it could add up quickly with many users. The system needs faster hardware or better algorithms to keep performance acceptable.

More Complexity

All four tactics make the system more complex. Intrusion detection needs a database of attack patterns that must be updated. Authentication needs password management and possibly connection to external authentication systems. Revoking access needs real-time blocking controls. Audit trails need secure storage and search capabilities.

Large Storage Needed

Audit trails create millions of log entries per day since the system handles 200 events per second (ASR-04). This requires expensive storage that grows as the system grows, affecting scalability and increasing operational costs.

Harder Integration

Authentication requires all three branch systems (CPS, RPS, BBS) to use compatible login methods. Since these systems use different approaches (CN-2), adding authentication requires significant work to modify old systems. This delays deployment and conflicts with BG-03's goal of responding quickly to market changes.

More Administrative Work

Managing security policies, reviewing logs, and responding to alerts requires constant administrator attention. Which increases costs because additional security staff may be needed.

- END -